

File permissions in Linux

Project description

In this project, I used Linux commands to inspect and manage file and directory permissions. I checked file permissions, interpreted permission strings, and modified permissions for files, hidden files, and directories to follow security best practices.

Check file and directory details

I used the `ls -la` command to display detailed information about files and directories, including hidden files. This command showed the file permissions, ownership, group, size, and modification date for each file and directory in the projects directory.

Describe the permissions string

`-rw-rw-r--`

The first character (-) indicates that the item is a regular file. The next three characters (rw-) show the owner's permissions, which allow reading and writing. The following three characters (rw-) show the group's permissions, which also allow reading and writing. The last three characters (r- -) show the other users can only read the file.

Change file permissions

I used the command `chmod o-w projects_k.txt` to remove write permissions from `-rw-rw-rw-` to `-rw-rw-r--`, ensuring that other users can read the file but cannot modify it.

Change file permissions on a hidden file

I used the command `chmod g-w,u-w,g+r .project_x.txt` to remove write permissions from `-rw- -w- - - -` to `-r- - -r - - -`, ensuring that no one has write permissions, and the user and group can read the file.

Change directory permissions

I used the command `chmod g-x drafts` to remove execute permission from the group on the `drafts` directory. This changed the permissions from `drwx- -x- - -` to `drwx- - - - -`, ensuring that only the owner (`researcher2`) can access the directory and its contents.

Summary

In this activity, I used Linux commands to review and modify file and directory permissions. I applied the principle of least privilege by removing unnecessary write and execute permissions while preserving the access required for authorized users. This project demonstrates my ability to manage Linux file permissions and secure system resources.